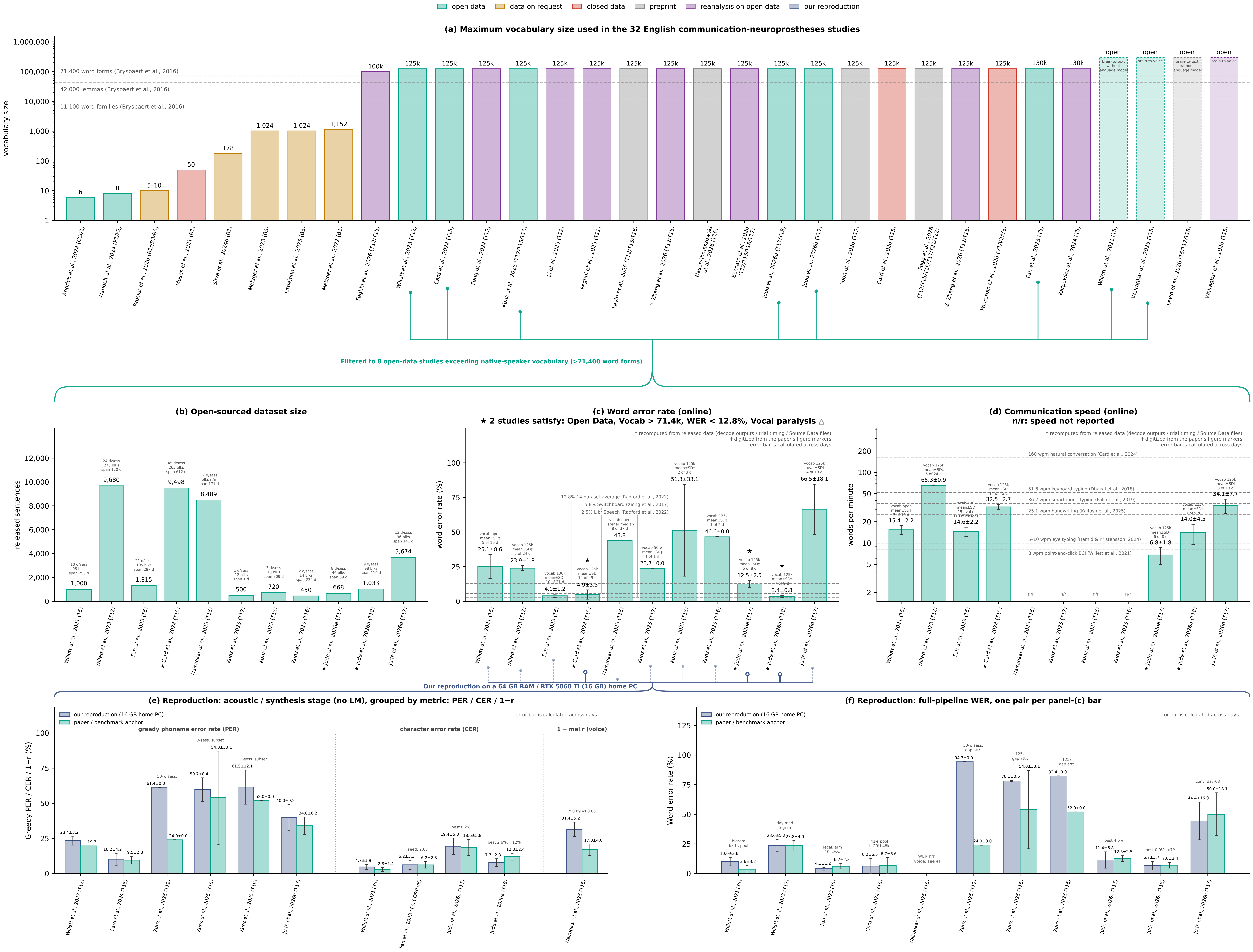
open data
data on request
closed data
preprint
reanalysis on open data
our reproduction
(a) Maximum vocabulary size used in the 32 English communication-neuroprostheses studies
vocabulary size
1,000,000
100,000
10,000
1,000
100
10
1
71,400 word forms (Brysbaert et al., 2016)
42,000 lemmas (Brysbaert et al., 2016)
11,100 word families (Brysbaert et al., 2016)
6
8
5–10
50
178
1,024
1,024
1,152
100k
125k
130k
open
brain-to-text without language model
brain-to-voice
Angrick et al., 2024 (CC01)
Wandelt et al., 2024 (P1/P2)
Brosler et al., 2026 (B1r/B3/B6)
Moses et al., 2021 (B1)
Silva et al., 2024b (B1)
Metzger et al., 2023 (B3)
Littlejohn et al., 2025 (B3)
Metzger et al., 2022 (B1)
Feghhi et al., 2026 (T12/T15)
Willett et al., 2023 (T12)
Card et al., 2024 (T15)
Feng et al., 2024 (T12)
Kunz et al., 2025 (T12/T15/T16)
Li et al., 2025 (T12)
Feghhi et al., 2025 (T12)
Levin et al., 2026 (T12/T15/T16)
Y. Zhang et al., 2026 (T12/T15)
Nason-Tomaszewski et al., 2026 (T16)
Boccato et al., 2026 (T12/T15/T16/T17)
Jude et al., 2026a (T17/T18)
Jude et al., 2026b (T17)
Yoon et al., 2026 (T12)
Card et al., 2026 (T15)
Fogg et al., 2026 (T12/T15/T16/T17/T21/T22)
Z. Zhang et al., 2026 (T12/T15)
Pouratian et al., 2026 (V1/V2/V3)
Fan et al., 2023 (T5)
Karpowicz et al., 2024 (T5)
Willett et al., 2021 (T5)
Wairagkar et al., 2025 (T15)
Levin et al., 2026 (T5/T12/T18)
Wairagkar et al., 2026 (T15)
Filtered to 8 open-data studies exceeding native-speaker vocabulary (>71,400 word forms)
(b) Open-sourced dataset size
released sentences
12,000
10,000
8,000
6,000
4,000
2,000
0
10 d/sess 95 blks span 253 d
1,000
24 d/sess 275 blks span 120 d
9,680
21 d/sess 105 blks span 287 d
1,315
45 d/sess 265 blks span 612 d
9,498
37 d/sess blks n/a span 171 d
8,489
1 d/sess 12 blks span 1 d
500
3 d/sess 18 blks span 309 d
720
2 d/sess 14 blks span 234 d
450
8 d/sess 46 blks span 89 d
668
9 d/sess 98 blks span 119 d
1,033
13 d/sess 96 blks span 141 d
3,674
Willett et al., 2021 (T5)
Willett et al., 2023 (T12)
Fan et al., 2023 (T5)
★ Card et al., 2024 (T15)
Wairagkar et al., 2025 (T15)
Kunz et al., 2025 (T12)
Kunz et al., 2025 (T15)
Kunz et al., 2025 (T16)
★ Jude et al., 2026a (T17)
★ Jude et al., 2026a (T18)
Jude et al., 2026b (T17)
(c) Word error rate (online)
★ 2 studies satisfy: Open Data, Vocab > 71.4k, WER < 12.8%, Vocal paralysis △
† recomputed from released data (decode outputs / trial timing / Source Data files)
‡ digitized from the paper's figure markers
error bar is calculated across days
word error rate (%)
100
75
50
25
0
12.8% 14-dataset average (Radford et al., 2022)
5.8% Switchboard (Xiong et al., 2017)
2.5% LibriSpeech (Radford et al., 2022)
vocab open mean±SD† 5 of 10 d
25.1±8.6
vocab 125k mean±SD‡ 5 of 24 d
23.9±1.8
vocab 130k mean±SD† 10 of 21 d
4.0±1.2
★
vocab 125k mean±SD 14 of 45 d
4.9±3.3
vocab open listener median 8 of 37 d
43.8
vocab 50-w mean±SD† 1 of 1 d
23.7±0.0
vocab 125k mean±SD† 2 of 3 d
51.3±33.1
vocab 125k mean±SD† 1 of 2 d
46.6±0.0
★
vocab 125k mean±SD† 6 of 8 d
12.5±2.5
★
vocab 125k mean±SD† 7 of 9 d
3.4±0.8
vocab 125k mean±SD† 4 of 13 d
66.5±18.1
(d) Communication speed (online)
n/r: speed not reported
† recomputed from released data (decode outputs / trial timing / Source Data files)
‡ digitized from the paper's figure markers
error bar is calculated across days
words per minute
200
100
50
20
10
5
2
160 wpm natural conversation (Card et al., 2024)
51.6 wpm keyboard typing (Dhakal et al., 2018)
36.2 wpm smartphone typing (Palin et al., 2019)
25.1 wpm handwriting (Kaifosh et al., 2025)
5–10 wpm eye typing (Hamid & Kristensson, 2024)
8 wpm point-and-click BCI (Willett et al., 2021)
vocab open mean±SD† 5 of 10 d
15.4±2.2
vocab 125k mean±SD‡ 5 of 24 d
65.3±0.9
vocab 130k mean±SD 15 eval d (10 released)
14.6±2.2
vocab 125k mean±SD 14 of 45 d
32.5±2.7
n/r
n/r
n/r
n/r
vocab 125k mean±SD† 6 of 8 d
6.8±1.8
vocab 125k mean±SD† 7 of 9 d
14.0±4.5
vocab 125k mean±SD† 8 of 13 d
34.1±7.7
Our reproduction on a 64 GB RAM / RTX 5060 Ti (16 GB) home PC
(e) Reproduction: acoustic / synthesis stage (no LM), grouped by metric: PER / CER / 1−r
our reproduction (16 GB home PC)
paper / benchmark anchor
error bar is calculated across days
Greedy PER / CER / 1−r (%)
greedy phoneme error rate (PER)
character error rate (CER)
1 − mel r (voice)
23.4±3.2
19.7
10.2±4.2
9.5±2.8
50-w sess.
61.4±0.0
24.0±0.0
3-sess. subset
59.7±8.4
54.0±33.1
2-sess. subset
61.5±12.1
52.0±0.0
40.0±9.2
34.0±6.2
4.7±1.9
2.8±1.4
seed: 2.65
6.2±3.3
6.2±2.3
best 8.2%
19.4±5.8
18.6±5.8
best 2.6%; <12%
7.7±2.8
12.0±2.4
r: 0.69 vs 0.83
31.4±5.2
17.0±4.0
Willett et al., 2023 (T12)
Card et al., 2024 (T15)
Kunz et al., 2025 (T12)
Kunz et al., 2025 (T15)
Kunz et al., 2025 (T16)
Jude et al., 2026b (T17)
Willett et al., 2021 (T5)
Fan et al., 2023 (T5, CORP v6)
Jude et al., 2026a (T17)
Jude et al., 2026a (T18)
Wairagkar et al., 2025 (T15)
(f) Reproduction: full-pipeline WER, one pair per panel-(c) bar
our reproduction (16 GB home PC)
paper / benchmark anchor
error bar is calculated across days
Word error rate (%)
125
100
75
50
25
0
bigram 63-tr. pool
10.0±3.6
3.6±3.2
day med. 5-gram
23.6±5.2
23.8±4.0
recal. arm 10 sess.
4.1±1.2
6.2±2.3
41-s pool biGRU-48k
6.2±6.5
6.7±6.6
WER n/r (voice; see e)
50-w sess. gap attr.
94.3±0.0
24.0±0.0
125k gap attr.
78.1±0.6
54.0±33.1
125k gap attr.
82.4±0.0
52.0±0.0
best 4.6%
11.4±6.8
12.5±2.5
best 0.0%; <7%
6.7±3.7
7.0±2.4
conv. day-68
44.4±16.0
50.0±18.1
Willett et al., 2021 (T5)
Willett et al., 2023 (T12)
Fan et al., 2023 (T5)
Card et al., 2024 (T15)
Wairagkar et al., 2025 (T15)
Kunz et al., 2025 (T12)
Kunz et al., 2025 (T15)
Kunz et al., 2025 (T16)
Jude et al., 2026a (T17)
Jude et al., 2026a (T18)
Jude et al., 2026b (T17)